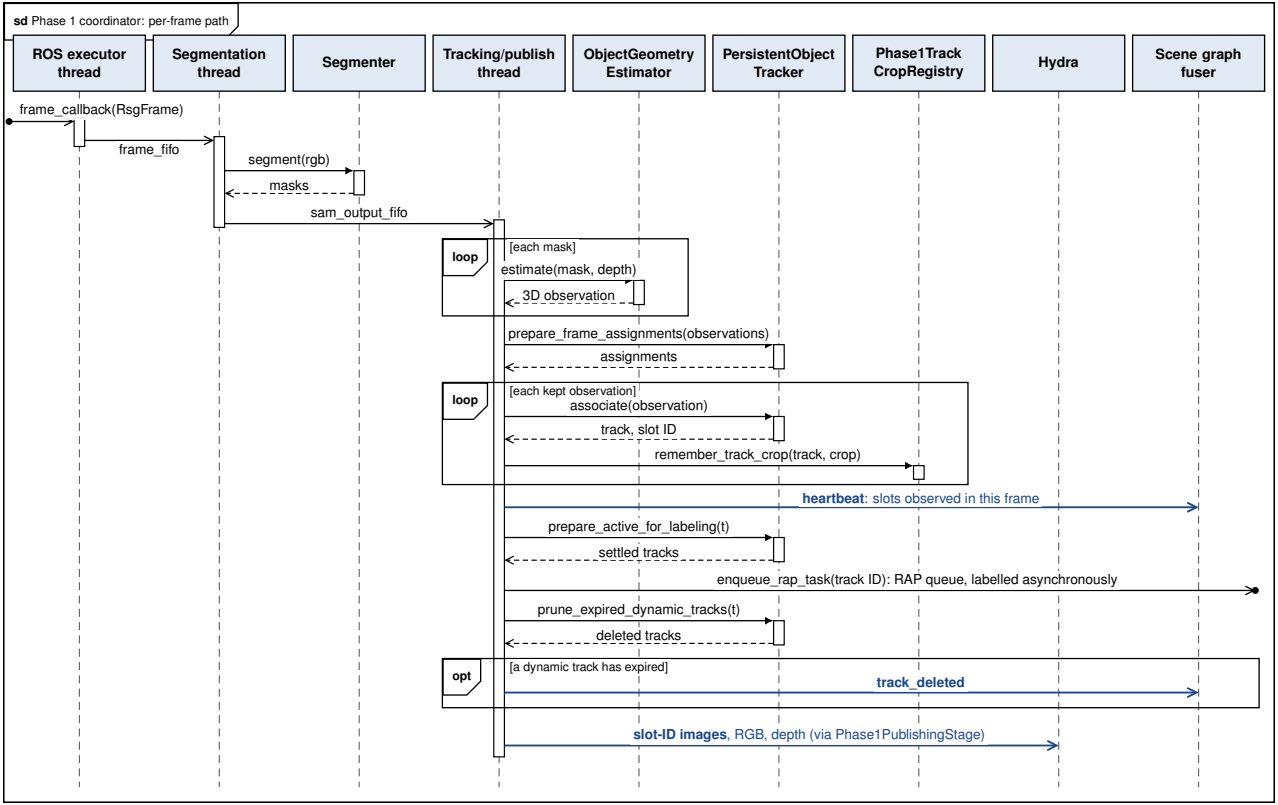
sd Phase 1 coordinator: per-frame path
ROS executor thread
Segmentation thread
Segmenter
Tracking/publish thread
ObjectGeometry Estimator
PersistentObject Tracker
Phase1Track CropRegistry
Hydra
Scene graph fuser
frame_callback(RsgFrame)
frame_fifo
segment(rgb)
masks
sam_output_fifo
loop
[each mask]
estimate(mask, depth)
3D observation
prepare_frame_assignments(observations)
assignments
loop
[each kept observation]
associate(observation)
track, slot ID
remember_track_crop(track, crop)
heartbeat: slots observed in this frame
prepare_active_for_labeling(t)
settled tracks
enqueue_rap_task(track ID): RAP queue, labelled asynchronously
prune_expired_dynamic_tracks(t)
deleted tracks
opt
[a dynamic track has expired]
track_deleted
slot-ID images, RGB, depth (via Phase1PublishingStage)